



LoRA: Low-Rank Adaptation of Large Language Models : Rank Ablation Re-implementation Study

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Introduction

PROBLEM: Fine-tuning large language models requires updating all parameters

- LoRA injects trainable low-rank matrices into frozen Transformer layers (no inference latency)

$$\mathbf{h} = \mathbf{W}_0 \mathbf{x} + \mathbf{B} \mathbf{A} \mathbf{x}$$

$\mathbf{W}_0$ frozen · $\mathbf{B} \in \mathbb{R}^{d \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times k}$ trainable · $r \ll \min(d, k)$

*A and B are initialized so $\Delta W = 0$ at the start of training.
At inference, they merge into W_0 with no overhead.*

Methodology

roberta-base (125M) used in place of GPT-3 (175B) due to compute constraints

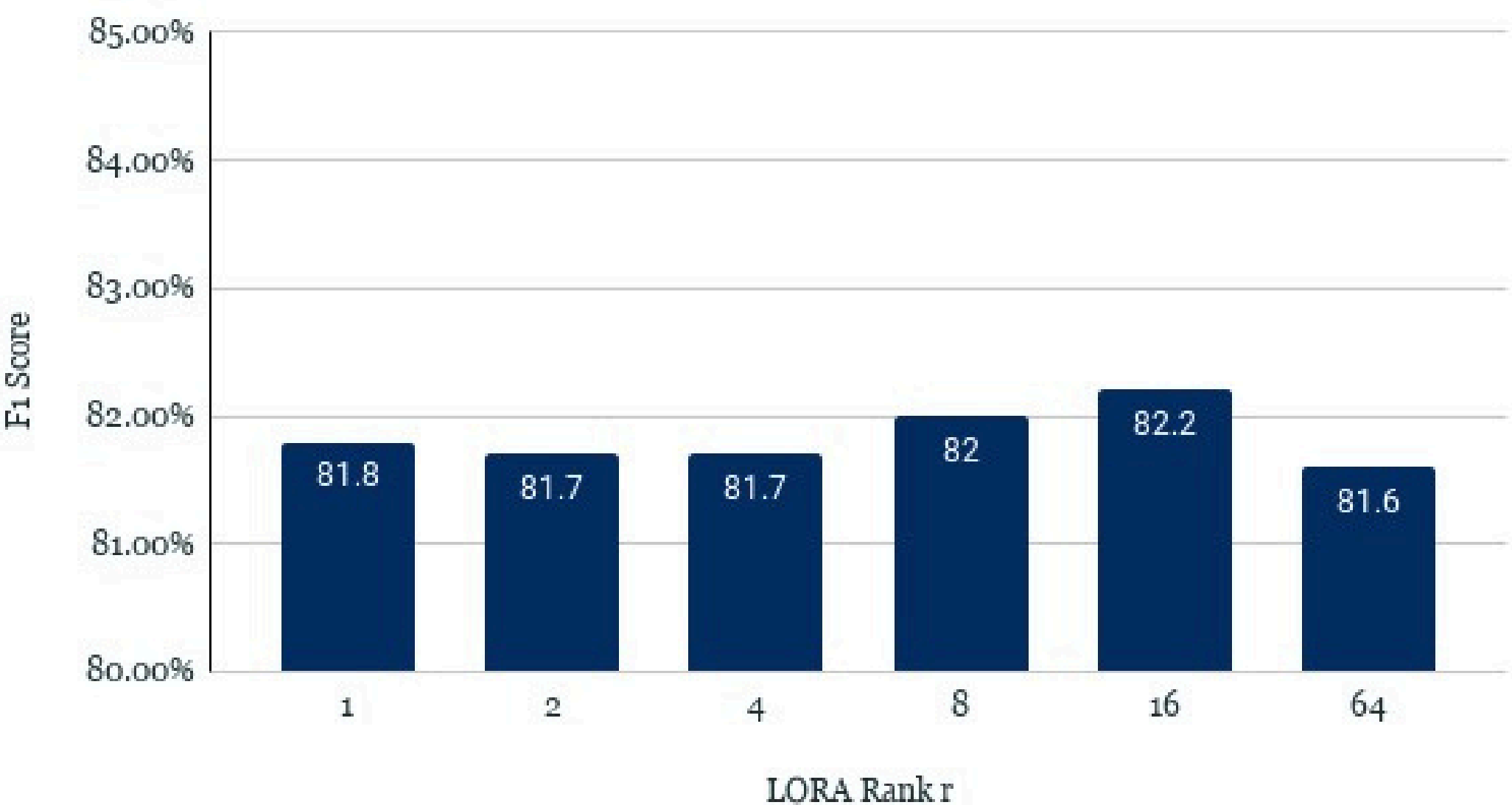
Parameter	Value
Base model	roberta-base (125M)
LoRA target	Wq, Wv
Dropout	0.1
Learning rate	3×10^{-4}
Batch size	32
Epochs	1
Ranks tested	1, 2, 4, 8, 16, 64

Conclusion

- Even with a much smaller model and limited compute, the rank ablation trend replicates at a fraction of the cost
- Weight updates during fine-tuning have a low intrinsic rank, meaning a single researcher can fine-tune a 125M parameter model in hours on CPU

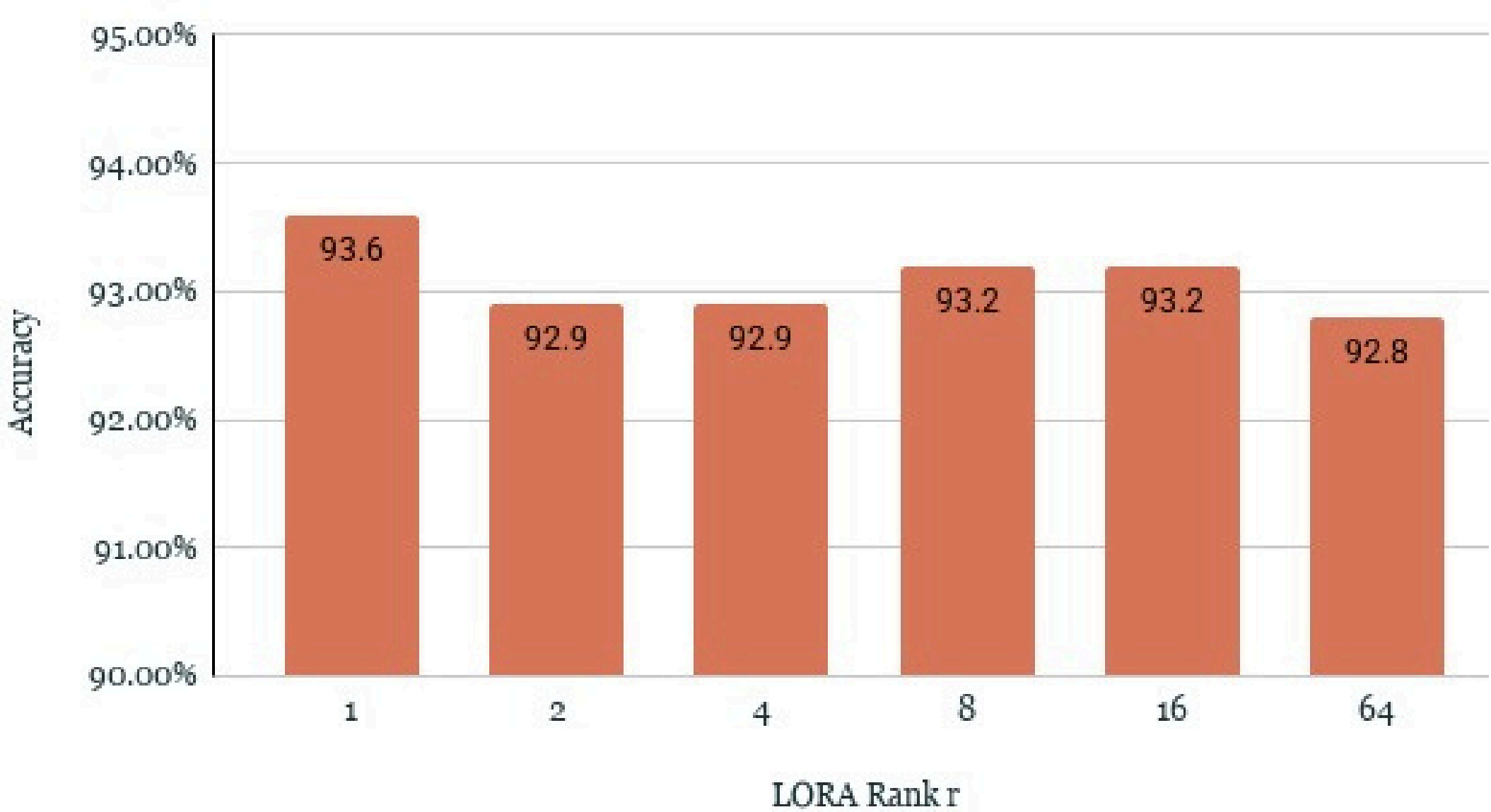
Results

MRPC — F1 Score vs. Rank



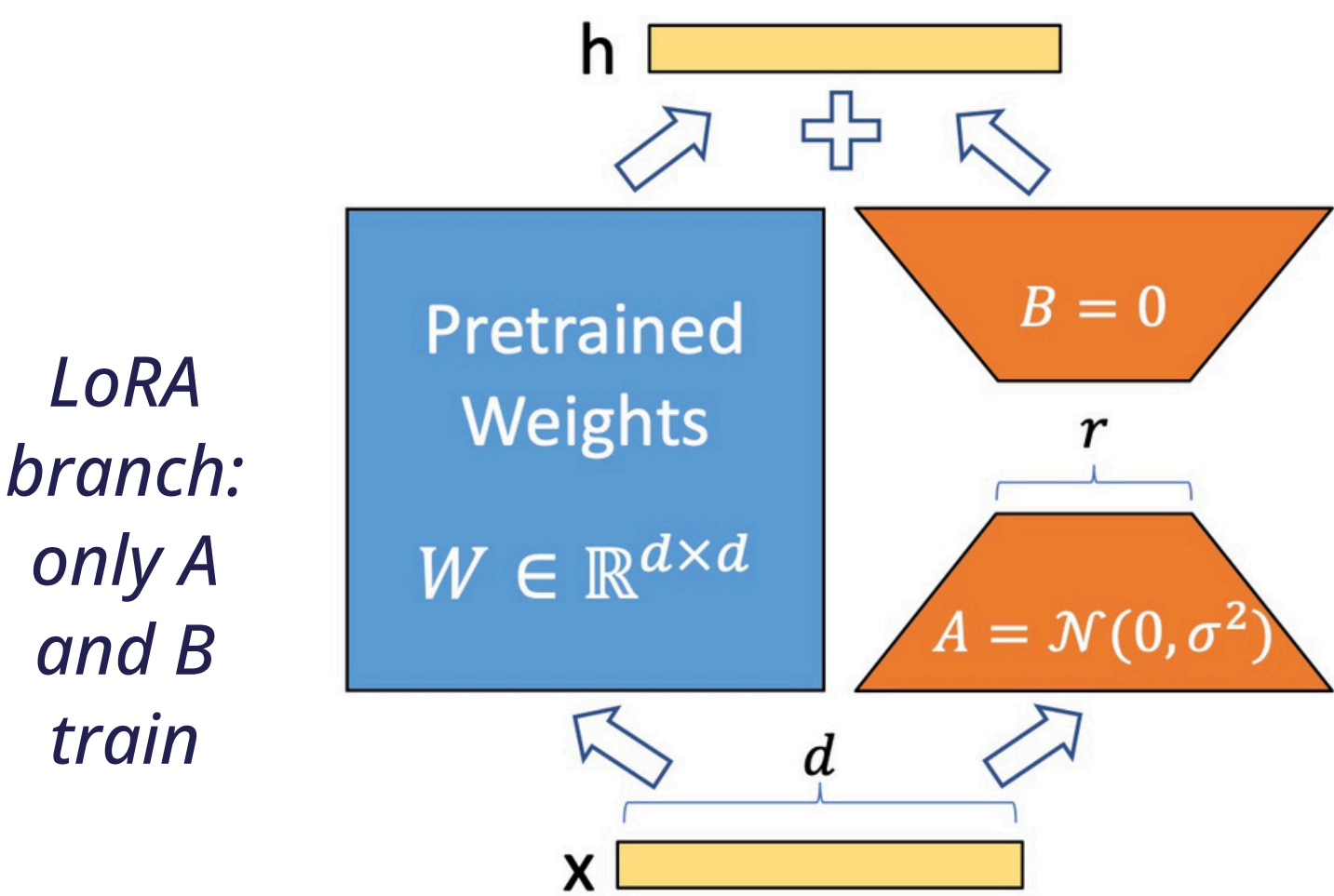
- SST-2 varies only 0.8% across all ranks
- r=64 uses 5× more params yet underperforms r=1 (92.8% vs. 93.6%)
- Our absolute scores are lower than the paper's (roberta-base vs. GPT-3, 1 epoch vs. multi-epoch, no MNLI warm-start), but the trend replicates cleanly.

SST2 — Accuracy vs. Rank



Observations

Rank r	SST-2 Acc.	MRPC F1	Params
r = 1	93.6% ★	81.80%	629K
r = 2	92.90%	81.70%	666K
r = 4	92.90%	81.70%	740K
r = 8	93.20%	82.00%	887K
r = 16	93.20%	82.2% ★	1.18M
r = 64	92.80%	81.60%	3.28M



Challenges

- Dependency conflicts required multiple debugging rounds before a single training run could complete.

Possible directions for future exploration:

- Train for more epochs to test if rank gaps widen with longer training
- Extend to GPT-2 or larger models to see if the same flat trend holds

References

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